Do Lagged Operational KPIs Improve Daily Revenue Forecasts? A Leakage-Safe Ablation on Online Retail Data

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***Abstract*—We test whether lagged operational key performance indicators (KPIs) improve daily retail revenue forecasts under a leakage-safe protocol. Using UCI Online Retail II, we compare ridge regression, random forest and LightGBM with revenue history and calendar features (Model A), and with additional lagged orders, active customers, average order value and cancellation rate (Model B). An expanding-window rolling-origin evaluation covers 305 target days per horizon using mean absolute scaled error (MASE). Corrected Diebold–Mariano tests assess significance at 1 and 7 days, with 28 days reported descriptively. All six machine-learning configurations significantly outperform seasonal naive under squared- and absolute-error loss. Against exponential smoothing (ETS), none is significant under the primary squared-error loss with protocol variance after Holm adjustment. Only random forest with Model A at 7 days is significant with Bartlett weights. Under absolute-error loss, 4 of 6 configurations outperform ETS at 1 day and all 6 at 7 days. Comparisons with SARIMA are descriptive. None of the 18 Model B–Model A comparisons across three targets detected a significant improvement. The post hoc feature-group ablation for revenue likewise detected none. Revenue MASE changes range from -3.2% to $+0.6\%$. Adding unavailable target-day KPIs reduces MASE to 0.16–0.24, demonstrating the effect of leakage.**

***Index Terms*—KPI forecasting, data leakage, rolling-origin evaluation, Diebold–Mariano test, retail analytics, SHAP**

I. Introduction

Revenue forecasts support inventory, staffing and budgeting decisions, and retail organizations increasingly produce them with machine learning (ML) models trained on transaction data [1]. Transaction data also yield operational KPIs: the number of orders, active customers, AOV and the cancellation rate. Because revenue equals orders times AOV, these KPIs are natural candidate predictors. The same identity makes them dangerous: if a same-period KPI enters the feature set, the model partly reconstructs the target instead of forecasting it. Such leakage produces accuracy estimates that cannot be reproduced at forecast time [2], and evaluation pitfalls of this kind are common in applied forecasting work [3].

This paper asks a narrow question: *when only information available at the forecast origin is used, do lagged operational KPIs improve daily revenue forecasts?* We answer it with a controlled ablation on a public dataset. Our contribution is methodological and deliberately modest:

- a leakage-safe forecasting protocol (expanding-window rolling origin, direct multi-horizon strategy, train-only preprocessing, automated leakage tests) with the main settings fixed in a configuration file before the backtest and later additions marked as post hoc;
- a controlled ablation of feature groups (revenue history and calendar vs. the same plus lagged driver KPIs) with Diebold–Mariano tests corrected for small samples and multiple comparisons;
- a leakage control experiment that quantifies how much same-period KPIs overstate accuracy.

We address three research questions. RQ1: are ML models (Ridge, random forest, LightGBM) significantly more accurate than the seasonal naive and ETS baselines for daily revenue at 1 and 7 days (SARIMA is compared with the seasonal naive baseline only)? RQ2: does adding lagged driver KPIs (Model B) significantly improve on revenue history and calendar alone (Model A) at 1 and 7 days? RQ3: how do accuracy and feature-group attributions change from 1 to 28 days (descriptively)?

II. Related Work

Large comparisons of statistical and ML forecasting methods showed that ML methods do not automatically outperform simple statistical baselines [4], while on tabular data tree ensembles remain strong compared with deep networks [5]. Retail forecasting has a long research tradition that stresses hierarchies, promotions and calendar effects [1]. The one-year predecessor of our dataset, Online Retail (December 2010 to December 2011), was introduced with a study of RFM-based customer segmentation [6], not with a time-series forecasting task. Arantes et al. [7] examined the trade-off between predictive accuracy and interpretability in daily retail sales forecasting using Rossmann store data.

Evaluation methodology is central to our design. Rolling-origin out-of-sample evaluation is the standard way to assess forecasts [8]; cross-validation for autoregressive models is

valid only under conditions discussed in [9]; and [3] list frequent pitfalls, including leakage and inappropriate error measures. We use the mean absolute scaled error (MASE), which is defined for series with zeros [10], and compare forecasts with the Diebold–Mariano (DM) test [11] with the small-sample correction of Harvey, Leybourne and Newbold (HLN) [12]. For explanation we use SHAP values [13].

A related study is a recent preprint by Kim, McCarthy and Lee [14], who forecast revenue jointly with its customer-base primitives (acquisition, repeat purchasing, spend per order) using weekly panels of 966 companies. Their multi-task model reduced mean total-sales error by 30% relative to the strongest representative established customer-base benchmark. Relative to a Transformer forecasting total sales directly, the reduction was 2.65%, with a non-significant paired difference ($p = .222$). Our study differs in scope: a single retailer at daily frequency, classical and tree-based models, and an explicit ablation of lagged driver KPIs under a leakage-safe protocol. As a preprint, [14] has not yet been peer reviewed.

III. Data and Protocol

A. Data and cleaning

We use UCI Online Retail II [15]: 1,067,371 transaction lines of a UK-based online retailer from 1 December 2009 to 9 December 2011. The two Excel sheets overlap from 1 to 9 December 2010 with 22,523 identical lines, which we keep only once. We further remove exact duplicates, bad-debt adjustments, lines with non-positive price, accounting, fee and test codes, postage lines, and two erroneous orders of more than 50,000 units that were cancelled within minutes. 1,021,346 lines remain. Cancellations (17,914 lines) are kept and used for the cancellation rate; lines without a customer identifier are kept for revenue and orders. The last day (9 December 2011) ends at 12:50 and is excluded. Every rule, with the number of affected lines and the revenue involved, is logged.

B. KPIs and calendar

Daily KPIs are computed on a full calendar of 738 days, of which 603 are trading days. Revenue is the gross merchandise revenue of non-cancelled lines; orders and active customers are distinct invoices and customer identifiers; AOV is revenue per order; the cancellation rate is the share of cancelled invoices. The retailer does not trade on Saturdays (one Saturday with sales in the whole period), on UK bank holidays and around Christmas. On the calendar series the autocorrelation at lag 7 is 0.64, and the in-sample MAE of the seasonal naive forecast is 10,599 with period 7, against 16,850 with period 6 and 15,830 for the one-day naive forecast. We therefore keep the 7-day calendar with zero revenue on non-trading days and use a seasonal period of $m = 7$.

C. Forecasting protocol

The test period runs from 7 February to 8 December 2011 (305 target days per horizon). It starts on the first Monday for which no forecast origin and no seasonal reference falls into the Christmas closure. For a target day T and horizon $h \in \{1, 7, 28\}$ the forecast origin is $T - h$ and the training set contains only targets observed up to $T - h$ (expanding window, direct strategy). All models are evaluated on exactly the same target days. Statistical models are re-estimated at every origin; ML models are re-estimated every seven days on all data up to the origin, while their lag features are always current. Hyperparameters are fixed in advance (no tuning on the test period); the Ridge penalty is selected by generalized cross-validation on the training window only. All settings were fixed in a configuration file before the backtest was run; the file was committed together with the first results, so the version history does not independently document this order. Analyses added after the main results (the no-Saturday tests, the feature-group ablation and the Bartlett sensitivity check) are marked as post hoc and logged in the configuration file.

D. Features

Model A uses revenue lags $h, \ldots, h + 6$ and the first four weekly lags not smaller than h, rolling means and standard deviations over 7 and 28 days of the series shifted by h, and calendar indicators (day of week, month, UK bank holiday, Christmas period), which are known in advance. Model B adds lags $h, \ldots, h + 6$ and a 7-day rolling mean (shifted by h) of orders, active customers, AOV and cancellation rate.

Leakage is tested automatically. Each lag must be at least h; same-period KPIs are forbidden; and a perturbation test overwrites all KPI values after $T - h$ with random noise and requires the features at T to be unchanged. These checks run in the unit tests and at every step of the backtest; a deliberately leaky feature set fails all three.

E. Models, metrics and tests

Baselines are the seasonal naive forecast ($m = 7$), additive ETS [16] and SARIMA$(1, 0, 1)(0, 1, 1)_7$ [17]. ML models are Ridge regression [18] with standardization, random forest [19] (300 trees, minimum leaf size 3) and LightGBM [20] (300 trees, learning rate 0.05, 15 leaves).

The primary metric is MASE [10]. Each error is scaled by the in-sample MAE of the seasonal naive forecast (lag 7) on the data available at its own origin; MASE is the mean of these scaled errors. We also report MAE, RMSE and WAPE. MAPE is not used because of zero-revenue days. The DM tests use unscaled forecast errors, with loss differentials defined by squared or absolute error. Scaling by the seasonal-naive denominator is applied only to MASE. Forecast accuracy is compared with the DM test [11] with the HLN correction [12], using autocovariances up to lag $h - 1$ with equal weights, squared-error loss (absolute-error loss as robustness) and Holm adjustment [21]. A Holm family is one target $\times$ horizon $\times$ comparison type $\times$ loss function; for Model B vs. Model A a family therefore holds 3 tests, one per ML model. Each model is compared with the seasonal naive forecast, each ML configuration with ETS, and Model B with Model A; ML models are not tested against SARIMA. Squared error is the protocol loss; because MASE is an absolute-error measure,

absolute-error results are reported alongside. Significance is interpreted for $h = 1$ and $h = 7$ only; p-values for $h = 28$ are computed and stored with the results but not interpreted, because the variance estimator is unreliable at this horizon (Section VI). As a post hoc sensitivity check, all tests are repeated with Bartlett weights $1 - k/h$.

IV. Experiments

The backtest covers three targets (revenue, orders, active customers), three horizons, six models and two feature sets, plus the leakage control. This paper focuses on revenue; orders and active customers are reported for the B-vs-A comparison only. The leakage control repeats Model B for Ridge and LightGBM and adds the values of orders, active customers, AOV, cancellation rate and quantity sold on the target day itself. Feature attributions are computed with TreeSHAP for every out-of-sample LightGBM forecast and aggregated by feature group; group permutation importance is computed as a cross-check. The code, experiment configurations, and result files are available at https://github.com/A2N19/retail-kpi-forecasting. The full backtest runs in under 7 minutes on a laptop CPU.

V. Results

A. RQ1: models vs. baselines

Table I reports MASE for daily revenue. With squared-error loss, all eight configurations (ETS, SARIMA and the six ML configurations) are significantly more accurate than the seasonal naive forecast at $h = 1$ and $h = 7$ (Holm-adjusted DM-HLN $p < 0.05$), also with Bartlett weights. With absolute-error loss, all six ML configurations remain significant, but ETS does not ($p = 0.058$ at $h = 1$, $p = 0.149$ at $h = 7$) and neither does SARIMA at $h = 7$ ($p = 0.053$). Random forest has the lowest MASE at both horizons: with Model A at $h = 1$ (0.752) and with Model B at $h = 7$ (0.756, ahead of Model A by less than 0.001), corresponding to a WAPE of about 0.28. The comparison with ETS depends on the loss. With absolute error, 4 of 6 ML configurations (Ridge and random forest, both feature sets) are significantly more accurate than ETS at $h = 1$, and 6 of 6 at $h = 7$ (Holm-adjusted). With squared error, only random forest with Model A at $h = 7$ is significant, and only with Bartlett weights; with the protocol variance no configuration is significant after the Holm adjustment, although 4 of 6 at $h = 1$ and 5 of 6 at $h = 7$ have unadjusted $p < 0.05$. Across all tests at $h = 1$ and $h = 7$, 7 of 204 decisions differ between the two variance estimators. ML models were not tested against SARIMA; the MASE values in Table I are descriptive for this comparison. At $h = 28$, reported descriptively, ETS and SARIMA have MASE 0.920 and 0.960 and all ML configurations stay below 0.83; the corresponding p-values are not interpreted (Section VI).

B. RQ2: lagged driver KPIs

Table II and Fig. 1 compare Model B with Model A at $h = 1$ and $h = 7$. For revenue the relative change in MASE lies between -3.2% and $+0.6\%$. No significant difference was detected in any of the 18 comparisons across the three targets, two horizons and three models, under either loss, with either variance estimator or without Saturdays. This is a null result: with 305 test days per horizon, our design did not detect a gain from lagged drivers; it does not show that such a gain cannot exist.

Ablation by driver group (post hoc). Table III adds one driver group at a time to Model A. Across all five columns of Table III, including All, MASE changes at 1 and 7 days range from -3.2% to $+3.1\%$. With the same Holm family definition (15 tests per family: five feature sets $\times$ three models), no group differs significantly from Model A under any loss or variance estimator. Recomputing Models A and B in the ablation reproduced the main forecasts to within 10^{-10}.

C. RQ3: horizons and feature attributions

Descriptively (Table I), from $h = 1$ to $h = 28$ the MASE of random forest (Model A) rises from 0.752 to 0.758 and that of Ridge (Model A) from 0.796 to 0.818, while LightGBM (Model A) falls from 0.849 to 0.787; seasonal naive rises from 0.981 to 1.065, ETS from 0.872 to 0.920 and SARIMA from 0.838 to 0.960. Fig. 2 shows the share of mean absolute SHAP values by feature group for LightGBM with Model B. The lagged driver KPIs receive 48%, 44% and 41% of the attribution at $h = 1$, 7 and 28, although no statistically significant accuracy gain from them was detected at $h = 1$ and $h = 7$. The calendar share grows with the horizon

TABLE I
Test-period MASE for daily revenue (305 target days per horizon). Set A: revenue history and calendar; set B: A plus lagged driver KPIs. Lower is better; best per horizon in bold (determined before rounding). Significance tests: see text (ML vs. SARIMA not tested); $h = 28$ is descriptive (p-values not interpreted, Section VI).

Model	Set	$h = 1$	$h = 7$	$h = 28$
Seasonal naive	–	0.981	0.980	1.065
ETS	–	0.872	0.915	0.920
SARIMA	–	0.838	0.861	0.960
Ridge	A	0.796	0.799	0.818
Ridge	B	0.792	0.804	0.826
Random forest	A	**0.752**	0.756	**0.758**
Random forest	B	0.755	**0.756**	0.769
LightGBM	A	0.849	0.799	0.787
LightGBM	B	0.833	0.773	0.785

TABLE II
Model B (with lagged driver KPIs) vs Model A for daily revenue. ΔMASE = (B$-$A)/A; p: DM-HLN, Holm-adjusted within each target $\times$ horizon $\times$ comparison type $\times$ loss family (3 tests: one per model); SE and AE loss.

h	Model	ΔMASE, %	p_{SE}	p_{AE}
1	Ridge	-0.5	1.000	1.000
1	Random forest	+0.3	1.000	1.000
1	LightGBM	-1.9	1.000	1.000
7	Ridge	+0.6	1.000	1.000
7	Random forest	0.0	1.000	1.000
7	LightGBM	-3.2	1.000	0.736

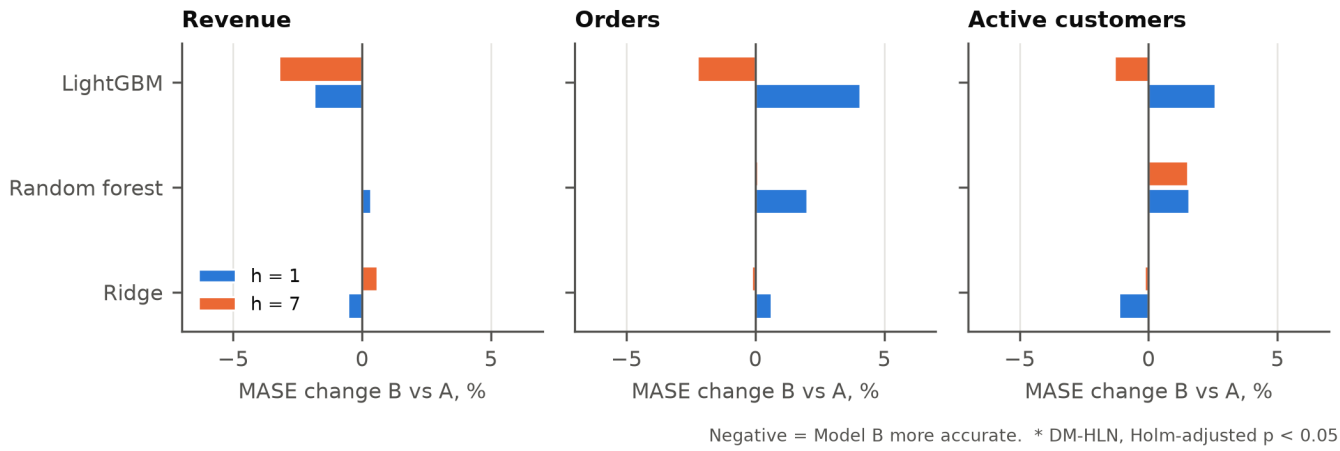


Fig. 1. Relative MASE change of Model B vs. Model A for each target, model and horizon $h \in \{1, 7\}$ (negative: Model B more accurate). No difference is significant (Holm-adjusted DM-HLN, $p \geq 0.05$).

TABLE III
FEATURE-GROUP ABLATION FOR DAILY REVENUE (POST HOC): MASE CHANGE (%) OF MODEL A PLUS ONE DRIVER GROUP (ORDERS, ACTIVE CUSTOMERS, AOV, CANCELLATION RATE), OR ALL FOUR (= MODEL B), RELATIVE TO MODEL A. NO DIFFERENCE IS SIGNIFICANT (HOLM-ADJUSTED DM-HLN; SE AND AE LOSS; PROTOCOL AND BARTLETT VARIANCE).

h	Model	Ord.	Cust.	AOV	Canc.	All
1	Ridge	-0.6	-0.6	-1.3	+0.2	-0.5
1	RF	-0.5	-1.1	+0.6	+2.8	+0.3
1	LGBM	-3.0	-1.1	-3.1	+3.1	-1.9
7	Ridge	-0.3	+0.7	-1.5	+0.9	+0.6
7	RF	-0.3	-1.1	-1.3	+2.1	0.0
7	LGBM	+0.6	+1.2	-1.1	-2.9	-3.2

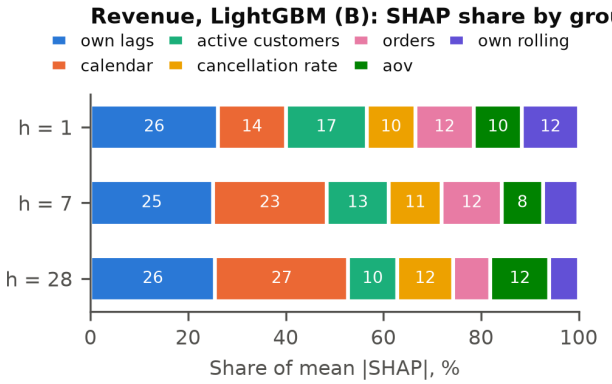


Fig. 2. Share of mean |SHAP| by feature group for out-of-sample LightGBM revenue forecasts (Model B). Shares sum to 100% before rounding; displayed values are rounded.

(14% to 27% in Model B; 26% to 39% in Model A). Group permutation importance ranks the calendar group first for both Ridge and LightGBM at all horizons.

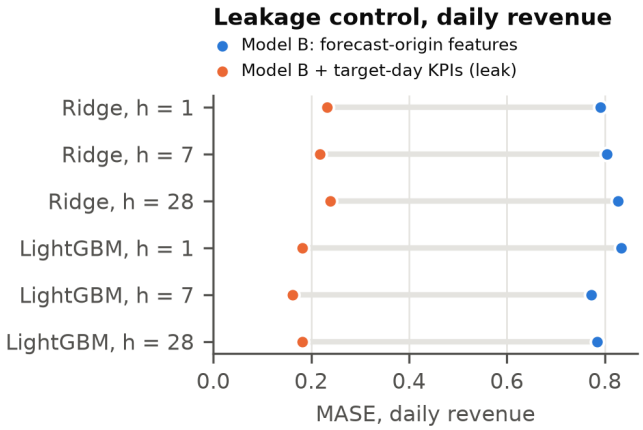


Fig. 3. MASE for daily revenue of Ridge and LightGBM with Model B features available at the forecast origin and with the target day's own orders, active customers, AOV, cancellation rate and quantity sold added (leakage); all horizons, descriptive.

D. Leakage control

Adding the target day's own KPIs reduces the MASE of Ridge and LightGBM from 0.77–0.83 to 0.16–0.24 (Fig. 3); the ratio of the legitimate to the leaky MASE is 3.4–4.8. For LightGBM at $h = 1$, WAPE falls from 0.311 to 0.068. The added target-day features are unavailable at the forecast origin, so such forecasts cannot be issued.

E. Robustness

Excluding Saturdays, on which revenue is almost always zero, raises all MASE values (seasonal naive: 1.142 at $h = 1$). Among Model A configurations at $h = 1$, Ridge (0.851) is slightly ahead of random forest (0.875), and the lowest MASE on this subset is Ridge with Model B (0.844). On this subset none of the 18 B-vs-A comparisons at $h = 1$ and $h = 7$ is significant under either loss; removing Saturdays

leaves gaps in the forecast sequence, so the DM variance is an approximation here.

VI. Discussion and Limitations

No statistically significant improvement from the lagged driver KPIs was detected in our setting, which is consistent with the non-significant gain reported by [14] for a very different design. One hypothesis is that at daily frequency the lagged drivers largely duplicate information already contained in lagged revenue, so the models redistribute attribution (Fig. 2) without gaining accuracy. We did not test this hypothesis directly.

The leakage control shows why protocol details matter: with the target day's KPIs a model would reach a MASE 3.4–4.8 times lower than the legitimate models, and such a result could easily be reported as a success.

Limitations: (i) a single online retailer whose sales are mostly domestic (United Kingdom: 85.4% of gross revenue and 92.0% of transaction lines), with one test year that includes one Q4 peak; (ii) Random forest and LightGBM used fixed hyperparameters; tuning inside each training window was not attempted. LightGBM is weaker than random forest and Ridge at $h = 1$ (MASE 0.849 vs. 0.752 and 0.796), so its ranking should not be generalized; (iii) ML models are re-estimated weekly rather than daily; (iv) identical transaction lines were treated as duplicates because the data contain no line numbers; (v) we did not estimate the minimum detectable effect; (vi) p-values for $h = 28$ are computed and stored with the results but not interpreted: with 27 equally weighted sample autocovariances and 305 forecasts, the protocol variance estimate is unstable. Among the 66 squared-error comparisons at this horizon (51 main comparisons for the three targets and 15 ablation comparisons for revenue), it is positive but below half of the lag-0 variance in 7 cases and negative in 1; with absolute error it is negative in 2 further ablation comparisons. Small positive variance estimates can inflate the magnitude of the test statistic; negative values trigger a fallback to the lag-0 variance. No fallback occurred at $h = 1$ or $h = 7$; a corrected estimator is left for future work; (vii) the feature-group ablation, the no-Saturday tests and the Bartlett check were added after the main results; (viii) deep learning models were not considered.

Model A's features are a subset of Model B's. The DM tests were not specifically calibrated for nested comparisons involving regularization and repeated estimation. Their non-rejection therefore does not establish that the additional KPIs contain no predictive information.

VII. Conclusion

Under a leakage-safe rolling-origin protocol, ML models forecast daily revenue of an online retailer significantly better than a seasonal naive baseline at 1 and 7 days under both squared and absolute error. Against ETS, under absolute error 4 of 6 ML configurations are significantly better at 1 day and 6 of 6 at 7 days; under squared error only random forest with Model A at 7 days is, and only with Bartlett weights. Adding lagged orders, active customers, AOV and cancellation rate together produced no statistically significant improvement at 1 and 7 days for any model and any of the three targets; for revenue, the post hoc ablation that adds one group at a time found no significant improvement either. Adding the target day's own KPIs, which are unavailable at the forecast origin, would lower MASE by a factor of 3.4–4.8. At 28 days the ML models have lower MASE than ETS and SARIMA, but the p-values at this horizon are not interpreted until a suitable variance estimator is used. We recommend that KPI forecasting studies report leakage controls together with feature-group ablations and significance tests whose variance estimator suits the horizon. Future work will replicate the protocol on other retail data and at weekly frequency.

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